PVsyst V8.1.8

PVsyst - Simulation report

Grid-Connected System

Project: Kampala_Marriott_Hotel

Variant: Kampala_Marriott_Hotel

No 3D scene defined, no shadings

System power: 185 kWp

Nsambya - Uganda

Author

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Project summary

Geographical Site
Nsambya
Uganda

Situation

Latitude	0.3042 °(N)
Longitude	32.5874 °(E)
Altitude	1178 m
Time zone	UTC+3

Project settings
Albedo 0.20

Weather data
Nsambya
Meteonorm 9.0 dll, Sat=100% - Synthetic

System summary

Grid-Connected System

No 3D scene defined, no shadings

Orientation #1
Fixed plane
Tilt/Azimuth 10 / 0 °

Near Shadings
no Shadings

User's needs
Unlimited load (grid)

System information

PV Array		Inverters	
Nb. of modules	299 units	Nb. of units	2 units
Pnom total	185 kWp	Total power	160 kWac
		Pnom ratio	1.16

Results summary

Simulation step	Hourly				
Produced Energy	291051 kWh/year	Specific production	1570 kWh/kWp/year	Perf. Ratio PR	85.44 %

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General parameters

Grid-Connected System — **No 3D scene defined, no shadings**

Simulation step
Hourly

Orientation #1
Fixed plane
Tilt/Azimuth 10 / 0 °

Models used
Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon
Free Horizon

Near Shadings
no Shadings

User's needs
Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer	Generic
Model	CHSM66RN(DG)/F-BH-620 (Original PVsyst database)
Unit Nom. Power	620 Wp
Number of PV modules	299 units
Nominal (STC)	185 kWp
Modules	23 units x 13 In series
At operating cond. (50°C)	
Pmpp	172 kWp
U mpp	491 V
I mpp	351 A

Inverter

Manufacturer	Generic
Model	80M1-HYA (Original PVsyst database)
Unit Nom. Power	80.0 kWac
Number of inverters	2 units
Total power	160 kWac
Operating voltage	160-1000 V
Max. power (=>40°C)	88.0 kWac
Pnom ratio (DC:AC)	1.16
Power sharing within this inverter	

Total PV power

Nominal (STC)	185 kWp
Total	299 modules
Module area	808 m²
Cell area	754 m²

Total inverter power

Total power	160 kWac
Max. power	176 kWac
Number of inverters	2 units
Pnom ratio	1.16

Array losses

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 23 mΩ
Loss Fraction 1.50 % at STC

Module Quality Loss
Loss Fraction -0.75 %

Module mismatch losses
Loss Fraction 2.00 % at MPP

Strings Mismatch loss
Loss Fraction 0.05 %

IAM loss factor
Incidence effect (IAM): Fresnel smooth glass, n = 1.526

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.998	0.981	0.948	0.862	0.776	0.636	0.402	0.000

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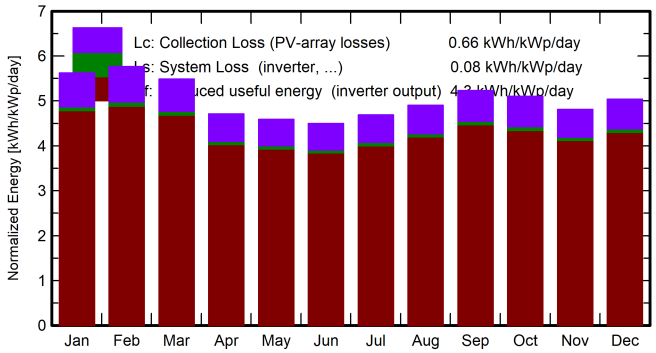
Main results

Simulation step
Hourly

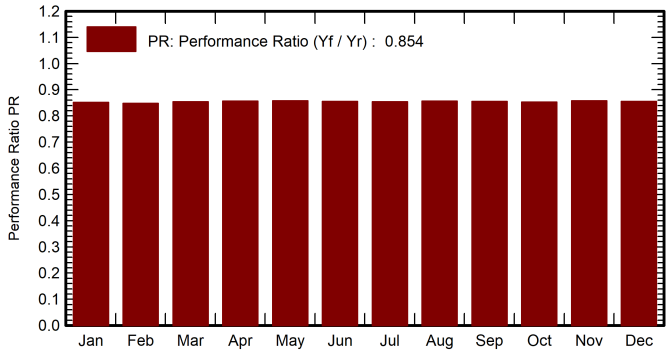
System Production

Produced Energy	291051 kWh/year	Specific production	1570 kWh/kWp/year
		Perf. Ratio PR	85.44 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

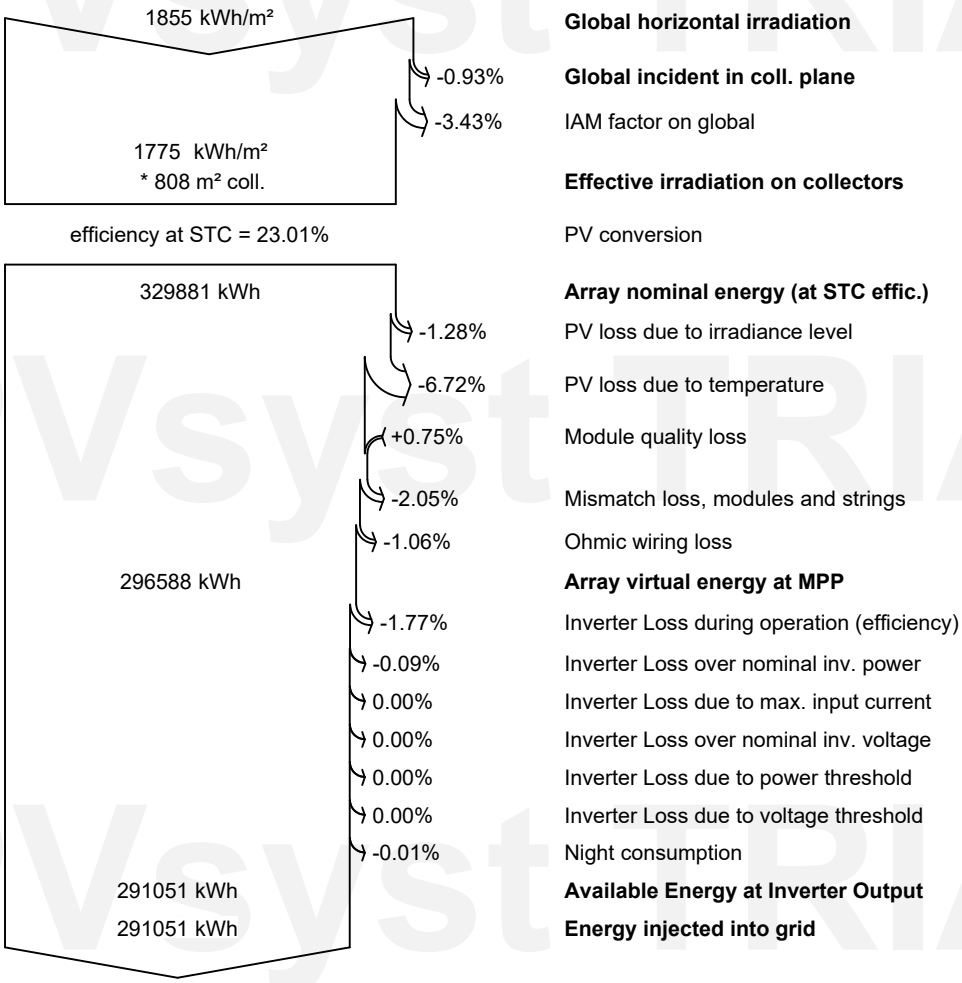
	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	163.9	65.15	23.02	174.4	169.0	28018	27523	0.851
February	155.8	60.45	23.52	161.5	157.0	25848	25387	0.848
March	170.3	76.10	23.21	170.2	164.9	27424	26942	0.854
April	146.6	72.32	22.79	141.3	136.1	22846	22443	0.857
May	152.4	71.94	22.59	142.3	136.5	23031	22628	0.858
June	147.4	61.35	22.19	135.0	129.0	21771	21387	0.855
July	158.5	56.87	21.80	145.4	139.4	23454	23038	0.855
August	160.2	63.78	21.90	152.1	146.5	24591	24154	0.857
September	159.5	65.84	22.11	157.0	152.1	25340	24885	0.855
October	155.5	68.89	22.31	158.1	153.4	25458	24993	0.853
November	138.1	73.98	22.30	144.3	139.6	23357	22933	0.857
December	146.8	67.95	22.42	156.1	151.3	25187	24738	0.855
Year	1855.0	804.62	22.51	1837.7	1774.7	296323	291051	0.854

Legends

GlobHor	Global horizontal irradiation
DiffHor	Horizontal diffuse irradiation
T_Amb	Ambient Temperature
GlobInc	Global incident in coll. plane
GlobEff	Effective Global, corr. for IAM and shadings
EArray	Effective energy at the output of the array
E_Grid	Energy injected into grid
PR	Performance Ratio

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Loss diagram



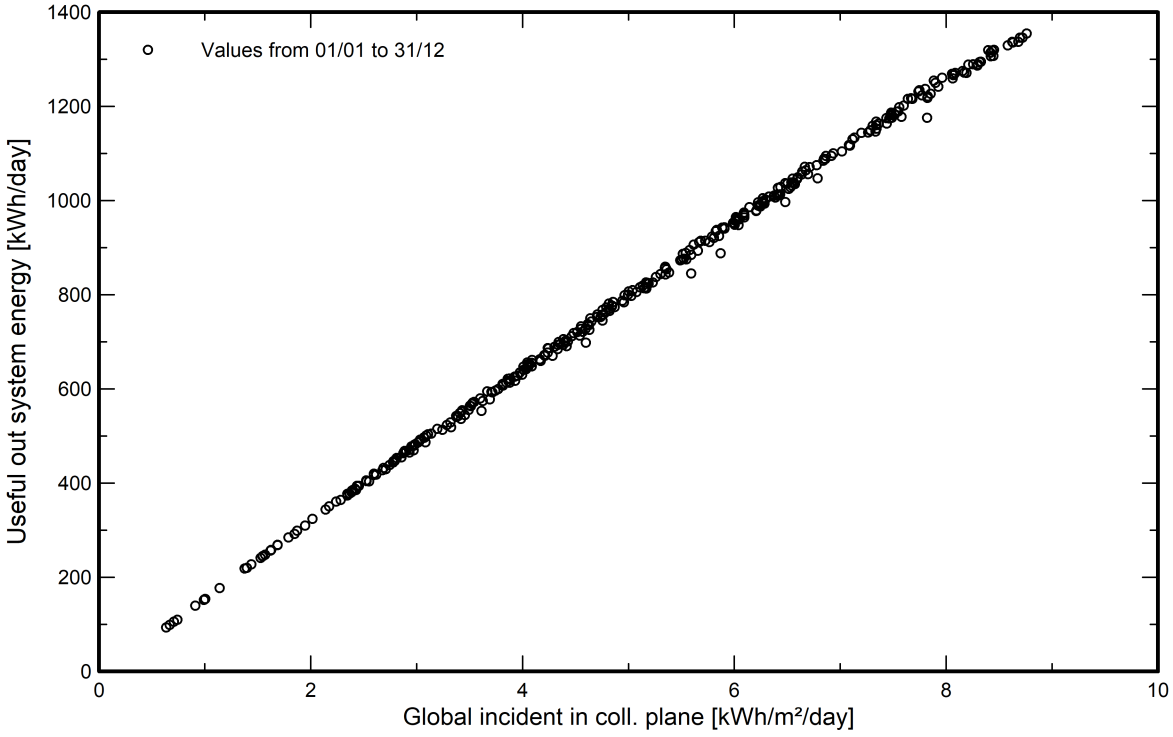
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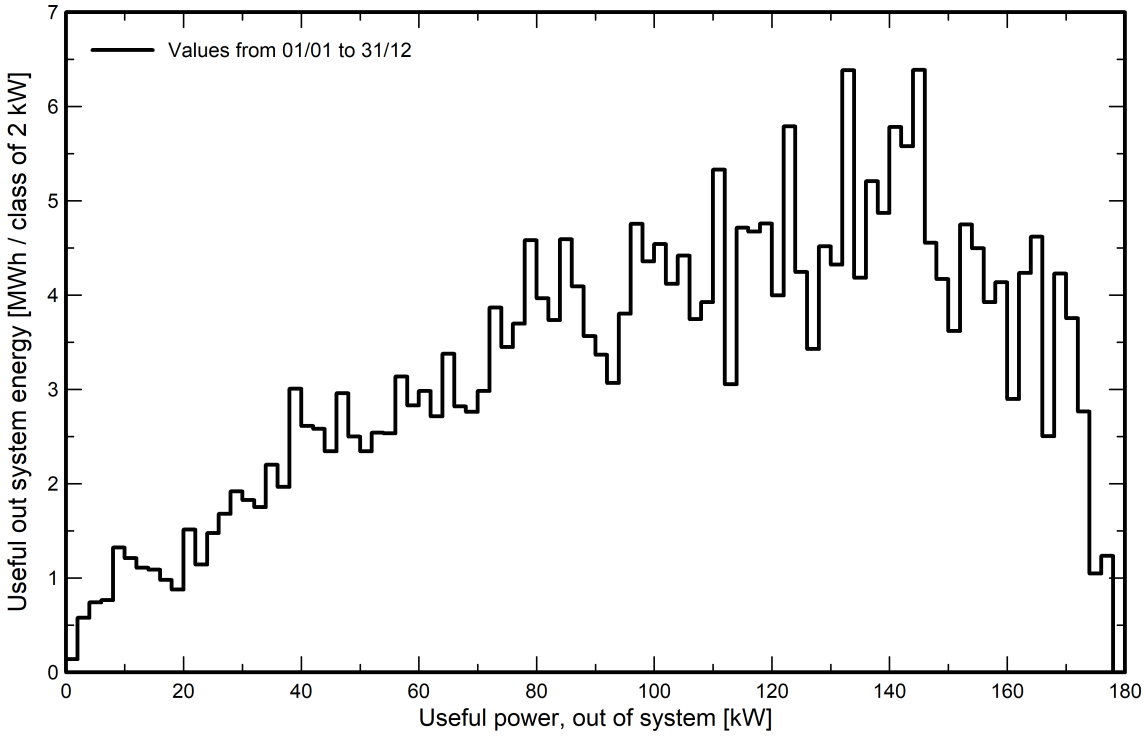
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Predef. graphs

Daily Input/Output diagram

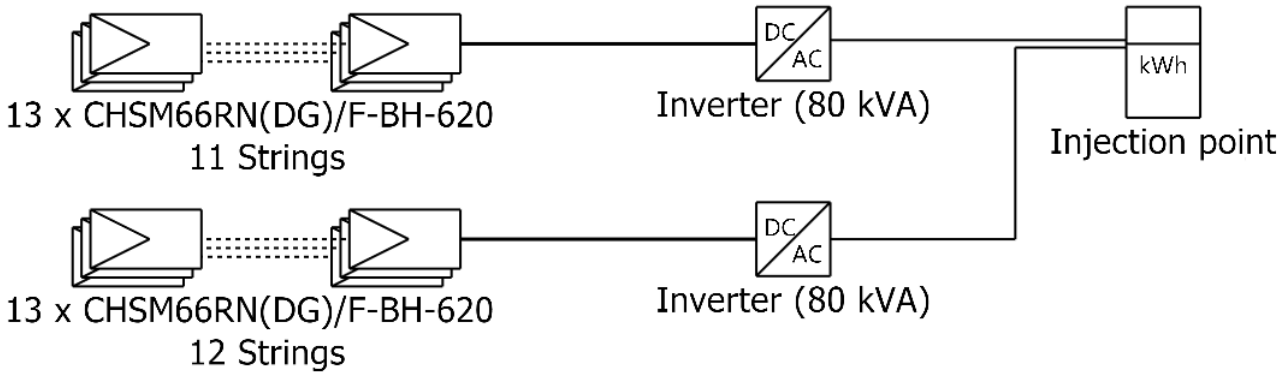


System Output Power Distribution



Single-line diagram

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PV module	CHSM66RN(DG)/F-BH-620
Inverter	80M1-HYA
String	13 x CHSM66RN(DG)/F-BH-620

	Nsambya_160_kW	
	VC0 : Nsambya_160_kW	09/11/26